

RAJ SHAH

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🔗 rajaxar.github.io/portfolio

Summary

Data scientist with 6+ years building statistical and machine learning models on real-world data. MS in Data Science and Public Policy from Carnegie Mellon. Works across causal inference, Bayesian hierarchical modeling, experimentation, and forecasting, with the Python and SQL to take models from prototype to production.

Work Experience

BlackCloak AI

Jan 2026 - Present

AI Engineer / Software Engineer

Remote

- Built internal tooling that automated the software development lifecycle, moving product pipelines from PRDs to shipped features in weeks rather than months; adopted by the 15-person US engineering team.
- Tech lead for the Home Blurring redesign, guiding engineers off a legacy monorepo to microservices and pub/sub.
- Built an adaptable insights platform so AI-generated summaries can be added across customer-facing services.

Aires

Jun 2025 - Jan 2026

Data Scientist

Pittsburgh, PA

- Developed Bayesian hierarchical models to estimate fixed and random effects driving customer survey outcomes.
- Modeled operational cash-flow time series with foundation forecasting models under limited historical data.
- Built LLM-powered auditing workflows with LangGraph and AWS Bedrock, reducing manual processing time by 85%.

Allianz Life

May 2024 - Apr 2025

Data Science Intern

Minneapolis, MN

- Built a full-stack ML/NLP solution (tokenization, TF-IDF) for legal-document triage, translating classifier outputs into prioritization workflows that reduced review time by ~40% and saved ~\$20K annually.
- Built an automated pipeline comparing 600+ permutations of models (BERT, LightGBM, XGBoost, scikit-learn) and feature-engineering methods, reaching 0.92 ROC-AUC while balancing accuracy, latency, and operational usability.
- Migrated dependencies for fraud-detection models worth \$4M+/quarter from legacy databases to the company standard using Python, SQL, and RStudio.
- Built Streamlit dashboards for analysts to auto-ingest data from multiple databases with multi-level recursive search.
- Prototyped a Cox proportional-hazards survival model to forecast policyholder lapses (exploratory analysis).

Ray Analytics (Startup)

Dec 2022 - Jul 2023

Founder / Chief Technology Officer

Washington, DC

- Founded and built a SaaS analytics platform for on-demand discourse analysis and policy-based influence identification, converting social/media engagement data into stakeholder-facing scores and reports.
- Trained, validated, and deployed XGBoost classifiers (94% ROC-AUC) on 12,000+ users from behavioral and engagement features to produce interpretable audience/influence scores.
- Used AWS API Gateway, Lambda, and React to build an automated backend and live dashboard serving 20+ customers.

Plainspoken Digital (Startup)

Mar 2022 - Jan 2023

Senior Software Engineer / Project Lead

Washington, DC

- Led migration from legacy data systems to Palantir Foundry, improving operational reporting efficiency by 2x.
- Designed automated ETL workflows (PySpark) integrating cloud and on-prem data into one analytics layer.
- Led Agile processes for a development team of 7 and managed intra-team project management.

Amazon Web Services (AWS)

Oct 2020 - Mar 2022

Software Development Engineer II (L5)

Crystal City, VA

- Led design and implementation of new features on the S3 Request Path, a codebase receiving billions of daily requests.
- Implemented S3 Bucket Keys, reducing encryption-related cost overhead by up to 99% for enterprise customers.
- Performed big-data latency analysis with Pandas/Seaborn; mentored engineers across the S3 Object Processing org.

Data Science for Social Good

May 2020 - Aug 2020

Data Scientist

Jacksonville, FL

- Led a team of three to analyze data for 6,000+ adult learners served by literacy services in Florida.

- Conducted causal analysis to evaluate program efficacy and identify key drivers of learner attrition, reducing dropout rates by over 20%.

Booz Allen Hamilton

Jun 2019 - Aug 2019

Data Scientist

Washington, DC

- Programmed a Convolutional Neural Network with TensorFlow reaching 88% precision, using transfer learning.

Education

Carnegie Mellon University, Heinz College

Pittsburgh, PA

Master of Science, Data Science and Public Policy

May 2025

- **Coursework:** Applied Econometrics, Machine Learning, Decision Analytics, Optimization, Labor Economics, Introduction to Deep Learning, Sports Analytics, Geographic Information Systems, Telling Stories with Data

University of Florida, College of Engineering

Gainesville, FL

Bachelor of Science, Computer Science, Minor in Mathematics

May 2020

- **Coursework:** Database Systems, Introduction to Data Science, Numerical Analysis, Algorithmic Design, Linear Algebra

Personal Projects

Outcome Evaluation of Pittsburgh Re-Entry Services | *Causal inference*

Jan 2025 - Apr 2025

- Led causal inference on 2,800+ participants to evaluate impact of services on employment, earnings, and recidivism.
- Designed quasi-experiments and causal inference frameworks to measure program impact; built 15+ year panel datasets.
- Quantified 3x higher odds of employment for matched graduates.

What's Driving MLB Batting Averages Down? | *Bayesian modeling*

Jan 2025 - Apr 2025

- Developed a Bayesian fixed-effects model using 240,000+ pitches to predict hit probability for causal trend analysis.
- Modeled random effects for batter, pitcher, year, and team with brms/Stana in R, partitioning variance in hitting.

Super Smash Bros. Melee Action-Value Analytics | *Production ML*

In Progress

- Built a win-probability value function predicting stock loss within an 8-second horizon; trained LightGBM on 10M+ frames at AUC 0.87, log-loss 0.27, Brier 0.082.
- Built a Bayesian hierarchical matchup model (PyMC) with crossed random effects over 60K+ games.
- Evaluated with GroupKFold by game to prevent frame leakage, plus calibration and probability-over-time plots.
- Benchmarked LightGBM against a 2-layer LSTM; lag-feature ablations showed snapshots miss trajectory data.

Fund the Vote: Reducing Voter Wait Times | *Predict-then-optimize*

Aug 2024 - Dec 2024

- Predicted local voting turnout (Census data, GeoPandas, LassoCV, hierarchical regression) and optimized precinct resource allocation with a Gurobi model to reduce wait times for marginalized voters.

Data Journalism: Survivor and the Data of Diversity | *Visualization*

- Built a scrollytelling investigation into how casting and player behavior shifted around diversity over 5 years, using sentence embeddings mapped onto BLS occupation/wage data.

Technical Skills

Modeling & Statistics: Causal Inference, Bayesian Hierarchical Modeling, Experiment Design, A/B Testing, Survival Analysis, Forecasting, Generalized Linear Models, Uncertainty Quantification, Econometrics, Optimization (Gurobi)

Machine Learning & Data Engineering: Python (Pandas, scikit-learn, PyTorch, statsmodels, NumPy), R (Stan/brms, R Shiny), SQL, LightGBM, XGBoost, BERT, NLP, Feature Engineering, Model Deployment, ETL/ELT, Spark/PySpark

Cloud & Infrastructure: AWS (Lambda, S3, API Gateway, Bedrock), GCP (Cloud Run, Pub/Sub), Azure, Palantir Foundry, Git, CI/CD

Software Engineering: Java, C++, JavaScript, TypeScript, React, GraphQL, REST APIs, design patterns, OOP, TDD, DevOps

Visualization & Communication: Tableau, Power BI, Streamlit, D3.js, Scrollama, GIS (ArcGIS, GeoPandas), Stakeholder Reporting, Technical Mentoring